

GeoSpectra — Architecture Documentation

Semantic Retrieval and Multi-Temporal Change Analysis of Satellite Imagery Smart India Hackathon 2026 · Problem Statement 26227 · Team QuantumCore (IIC_TMSL_P6_S21)

This document describes the complete, current architecture of GeoSpectra: the ingestion pipeline, semantic retrieval subsystem, multi-temporal change analysis, quality and false-alarm suppression, Sentinel-1 SAR evidence layer, temporal intelligence (velocity / acceleration / storyline), discovery clustering, the AOI Explorer geospatial map, the grounded on-device Qwen analyst assistant, the analyst evidence report, the human-in-the-loop review workflow with SHA-256 audit fingerprints, and the incremental onboarding pipeline — all operating fully offline on commodity CPU hardware.

Related documents: `PROJECT_WORKFLOW.md` (index build & ingestion procedure), `docs/EVALUATION.md` (reproducible evaluation report), `docs/PROVENANCE.md` (model & dataset provenance), `PERFORMANCE_REPORT.md` and `FINAL_VALIDATION_REPORT.md` (measured performance and validation evidence).

1. Executive Summary

GeoSpectra is an analyst-oriented, on-premises satellite intelligence platform built around real, locally stored satellite imagery. Its core capability is the combination of:

- **Semantic retrieval** over real satellite tiles using a shared image/text embedding space (RemoteCLIP ViT-B/32, 512-d), served by a FAISS vector index;
- **Multi-temporal change analysis** that fuses optical semantic drift with multi-spectral evidence under quality controls;
- **Sentinel-1 SAR** as independent, cloud-penetrating corroboration;
- **Temporal intelligence** — velocity, acceleration and storylines that turn pairwise detections into rates and trends;
- **Human-in-the-loop review** with a ranked queue, confirm/reject decisions, feedback-informed reranking, and a tamper-evident SHA-256 audit trail;
- **A grounded on-device Qwen 2.5 assistant** that answers from measured facts and refuses to over-claim, with a deterministic fallback;
- **One-click analyst evidence reports** (PDF/JSON) generated locally with full source-scene provenance;
- **Incremental onboarding** of new areas through a live, staged pipeline with duplicate detection — no index rebuild;
- **Complete offline operation**: no cloud services or external APIs are called after models and data are staged locally.

Every number the system displays is measured from the actual catalog. Where evidence is unavailable, the system says so.

2. Problem Definition and Objectives

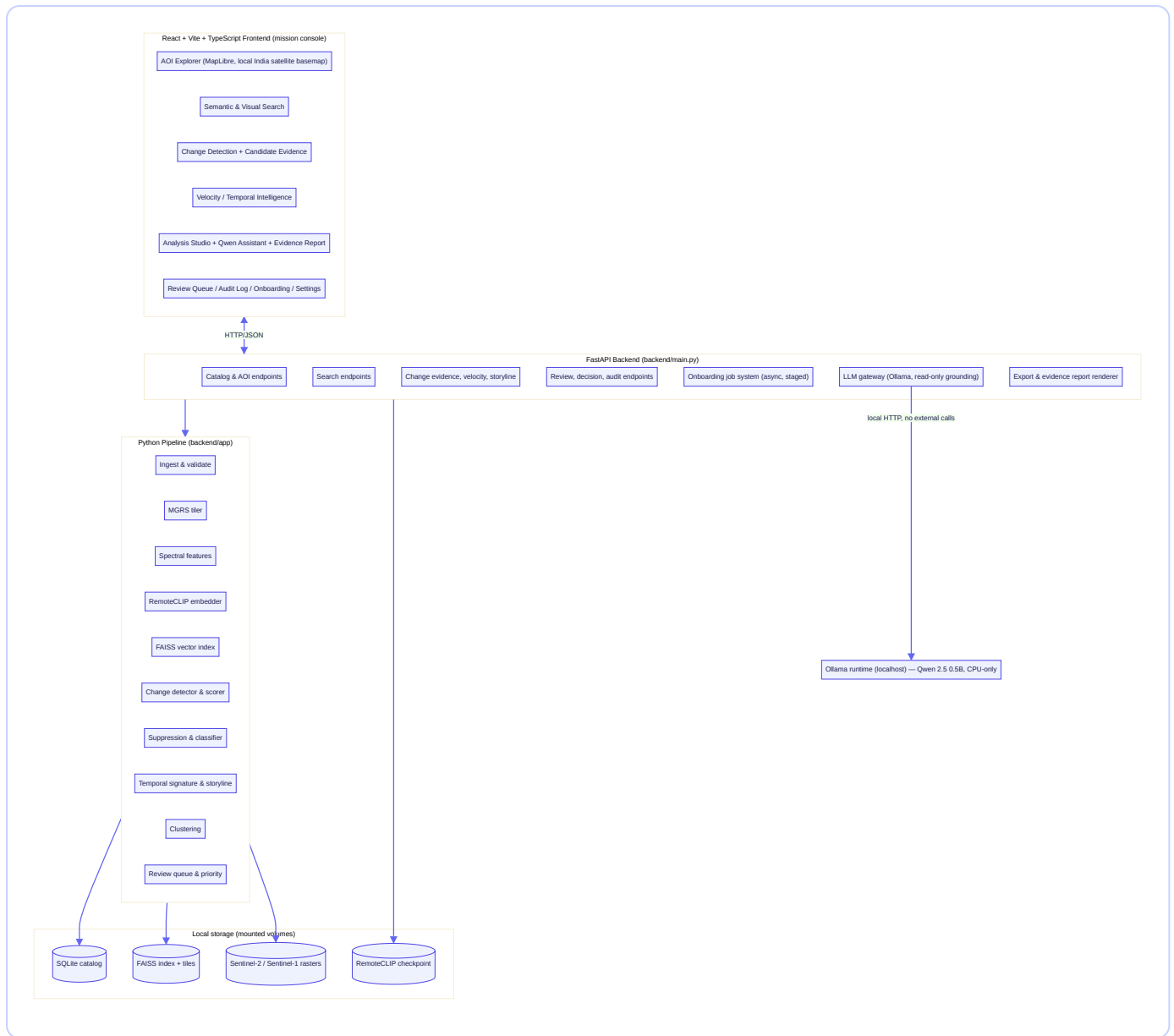
Earth-observation archives are expanding rapidly, but conventional catalogues are searchable only by metadata (coordinates, date, platform). Analysts must already know where and when to look. The problem statement requires an operational system that makes an archive queryable **by meaning** and **by change over time**, while remaining on-premises, incrementally ingestible, provenance-preserving, and robust to false change from season, atmosphere, viewing geometry, registration error and sensor differences.

GeoSpectra's objectives map one-to-one onto the six required capabilities:

PS capability	GeoSpectra implementation
2.2.1 Semantic & multimodal retrieval	/search/text + /search/image over RemoteCLIP + FAISS, ranked, AOI/date/K filters
2.2.2 Multi-temporal change analysis	hybrid scoring, zero-shot classification, earliest supported observation, velocity
2.2.3 False-alarm suppression	SCL/quality gates, calibrated thresholds, seasonal trajectories, SAR corroboration
2.2.4 Discovery & clustering	embedding-based clustering + similar-tile branching
2.2.5 Analyst workflow & provenance	review queue, confirm/reject + rationale, SHA-256 audit log, evidence reports, learner reranking
2.2.6 Scale, incremental ingestion, sovereignty	FAISS append, staged onboarding with dedup, offline enforced, GeoTIFF/COG

3. Overall System Architecture

The system comprises three layers, packaged as one CPU Docker container:



Sovereignty model: after the RemoteCLIP checkpoint, Qwen weights and imagery are staged locally, the system runs with network access disabled. Offline execution is enforced in the application settings — it is a stated runtime property, not an assumption.

4. Module Map (verified source tree)

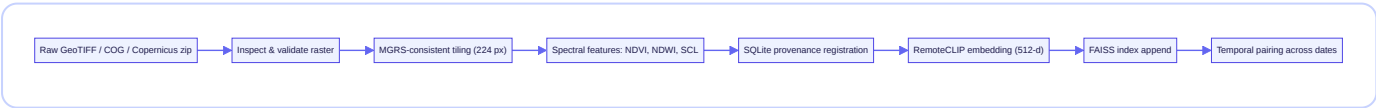
Module	Role
backend/app/pipeline/ingest.py	scene ingestion: validation, tiling, features, registration, embedding, indexing
backend/app/pipeline/batch_validate.py	batch validation of Copernicus archives / GeoTIFF folders
backend/app/pipeline/onboard_aoi.py	staged AOI onboarding: discovery → resolution → validation → ingestion → finalization, with per-scene and per-stage callbacks
backend/app/geospatial/reader.py	raster reading (GeoTIFF/COG), CRS/transform validation
backend/app/geospatial/tiler.py	MGRS-consistent tiling (224 px ≈ 2.24 km footprint at 10 m)
backend/app/geospatial/features.py	NDVI, NDWI, SCL/cloud/water quality features
backend/app/geospatial/catalog_db.py	SQLite catalog, provenance, record_decision with SHA-256 fingerprint
backend/app/geospatial/scene_stack.py	per-tile observation stacks
backend/app/geospatial/copernicus_metadata.py	Copernicus archive metadata extraction
backend/app/geospatial/rendering.py , sar_rendering.py	imagery & SAR rendering primitives
backend/app/embeddings/clip_embedder.py	RemoteCLIP ViT-B/32 embedding (512-d)
backend/app/index/vector_index.py	FAISS ID-mapped index; incremental append
backend/app/index/search.py	vector similarity search (text & image queries)
backend/app/index/remoteclip_migration.py	checkpoint staging/migration utilities
backend/app/change/detector.py	temporal pairing & hybrid change scoring
backend/app/change/adaptive.py	context-adaptive weighting of semantic vs spectral signals
backend/app/change/classifier.py	zero-shot change-type classification (CLIP)
backend/app/change/calibration.py	promotion-threshold calibration
backend/app/change/fusion.py	optical/SAR fusion with rescue guard
backend/app/change/sar.py	SAR change evidence branch
backend/app/change/heatmap.py	difference heatmaps / masks
backend/app/change/temporal_signature.py	velocity, acceleration, temporal signature classes
backend/app/change/storyline.py	classified storyline stages (short/seasonal)
backend/app/change/landcover.py	heuristic land-cover context

Module	Role
backend/app/change/priority.py	strategic priority (AOI tier, GeoJSON zones, hotspot proximity)
backend/app/change/reranker.py	review ordering (legacy-compatible helper; production ordering in review/queue.py)
backend/app/change/llm_brief.py	Ollama/Qwen grounded analyst brief
backend/app/change/analyst_chat.py	grounded multi-turn analyst chat
backend/app/change/brief.py, narrative.py	deterministic summaries & narratives
backend/app/change/evidence_report.py	analyst evidence report (PDF/JSON)
backend/app/discovery/clustering.py	embedding-based clustering of tiles
backend/app/review/queue.py	analyst triage & review queue ordering
backend/app/cli.py	operational CLI (ingest, cluster, velocity, SAR onboarding)
backend/main.py	FastAPI application (~50 endpoints, async onboarding jobs)
backend/rendering.py	thumbnails, mosaics, pixel-change maps, difference images

5. Technology Stack

Layer	Technology	Role
Satellite data	Sentinel-2 L2A, Sentinel-1 GRD (VV/VH)	real imagery, staged locally
Raster processing	Rasterio, NumPy	GeoTIFF/COG reading, masking, statistics
Spatial normalization	common CRS + MGRS-consistent tile identity	physical temporal correspondence
Optical features	NDVI, NDWI, SCL	vegetation/water context and quality gates
Semantic embeddings	RemoteCLIP ViT-B/32 (512-d)	shared image/text representation
Vector search	FAISS (ID-mapped, L2)	similarity retrieval, incremental append
Catalog	SQLite (additive migrations; PostgreSQL+PostGIS path documented)	provenance & metadata
On-device LLM	Qwen 2.5 0.5B via Ollama (CPU)	grounded analyst assistant
Backend	FastAPI (Python)	API layer, job system, export
Frontend	React 18, Vite, TypeScript, Tailwind CSS, Radix UI, TanStack Query, Recharts, MapLibre GL	mission-console analyst workspace
Geospatial UI	MapLibre GL + local India satellite basemap raster	offline AOI Explorer
Testing	pytest (regression suite), Playwright (browser suite)	pipeline & route verification
Packaging	Docker, docker-compose	single CPU container, mounted data/ + models/

6. Data Ingestion and Geospatial Preprocessing



- Inspect and validate** — readable raster structure, CRS, transform and resolution are checked before anything is accepted; invalid scenes are rejected with reasons.
- Tile** — scenes are split into fixed-size tiles on a shared spatial grid so that the same ground cell is comparable across dates and sensors.
- Features** — per-tile NDVI/NDWI and SCL-derived cloud/snow/shadow/valid pixel fractions are computed and stored.
- Register** — scene and tile provenance (source path, acquisition date, sensor, bounding box) is written to the SQLite catalog.
- Embed & index** — each tile's RGB representation is embedded with RemoteCLIP and appended to the FAISS index. The index is never rebuilt for new data; vectors are appended, keyed to catalog rows.

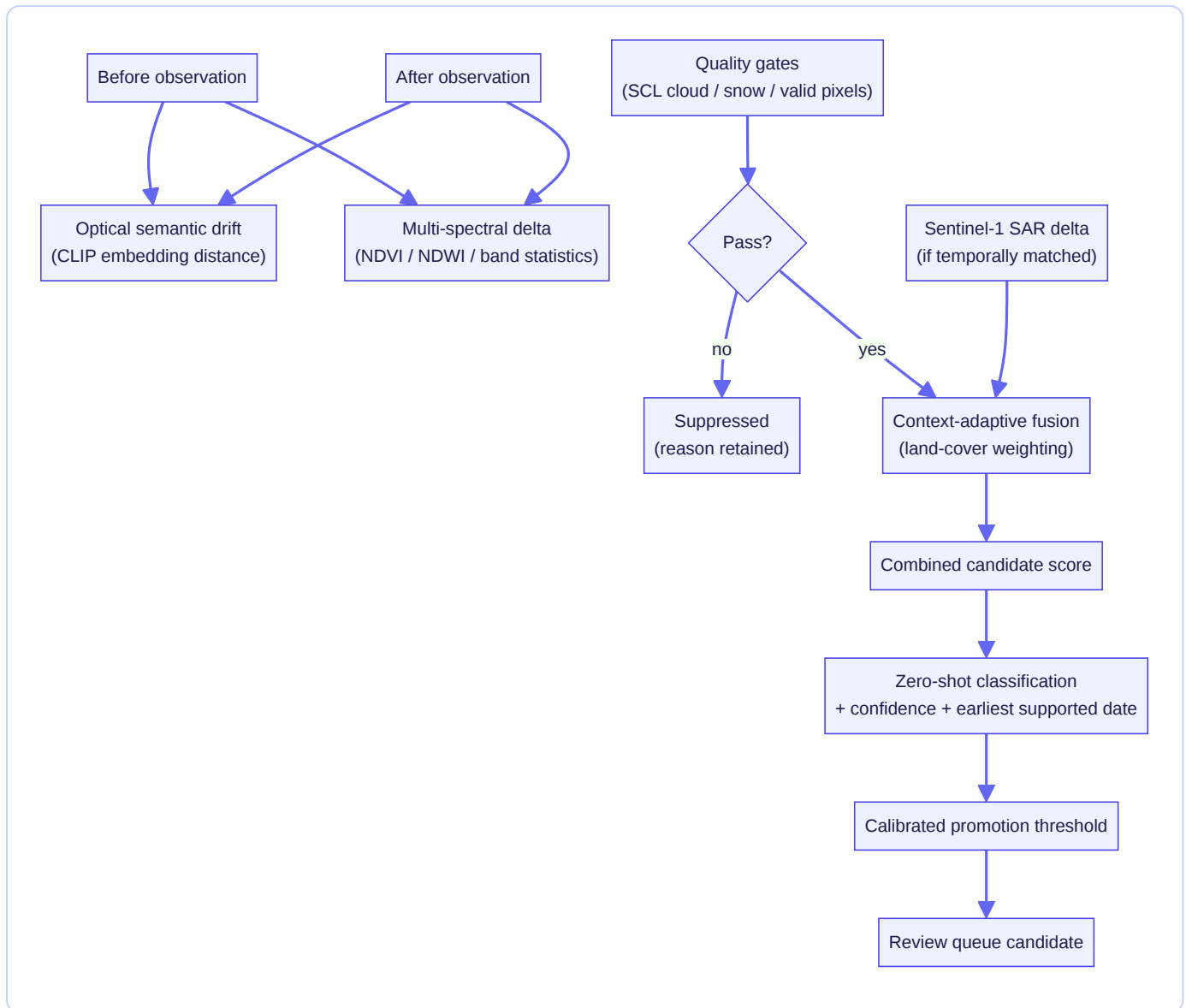
The semantic branch consumes an RGB-like representation; the full multispectral cube remains available to the physical-evidence and quality paths.

7. Semantic and Multimodal Retrieval

- **Text query search** (`POST /search/text`): the natural-language query is encoded by RemoteCLIP's text tower; FAISS retrieves the nearest tile vectors by cosine similarity. No keyword tags, manual labels or task-specific training are involved.
- **Image chip search** (`POST /search/image`): a 224×224 RGB reference chip is embedded in the same space; retrieval returns visually and semantically similar locations across the archive.
- **Ranking and filters**: results are rank-ordered by similarity and can be refined by AOI, date range and top-K, alongside conventional metadata filters.
- **Similar-tile branching**: `/tiles/{id}/similar` and before/after-tile similarity galleries let an analyst branch from any result to comparable locations — discovery without constructing a new query.

8. Multi-Temporal Change Analysis

For each ground cell, consecutive observations are paired and scored by a **hybrid signal**:



- **Semantic drift** captures appearance/disappearance/structural modification even when spectra shift subtly; **spectral delta** captures physical land-cover transitions.
- **Adaptive weighting** (`change/adaptive.py`) re-balances the two signals by land-cover context (e.g. urban/bare vs water).
- **Zero-shot classification** labels supported change types — construction, vegetation clearance, water-extent variation, road development — with an explicit confidence, and reports the **earliest supported observation** at which the change is backed by usable imagery.
- **Calibration** (`change/calibration.py`) sets the promotion threshold from data rather than a fixed magic number, favouring analytically useful precision over indiscriminate recall.

9. False-Alarm Suppression and Quality Handling

Seasonality, clouds, haze, snow, shadows, radiometric inconsistency and co-registration error are treated as confounds, not as change:

- SCL-based quality gates compute cloud/snow/valid-pixel fractions for **both** observations; failing candidates are suppressed before reaching the analyst.

- Per-candidate **historical NDVI/NDWI trajectories** are plotted across all ingested scenes for the tile, separating seasonal cycles from structural change.
- **SAR rescue guard** (verified in `FINAL_VALIDATION_REPORT.md`): a valid, temporally matched SAR observation with a low SAR score does **not** bypass optical quality suppression — SAR can corroborate, never smuggle through, a bad pair.
- Suppression is transparent: suppressed candidates remain inspectable with the reason attached (`source_imagery_unavailable_before`, cloud fraction gates, etc.).

On the current archive this discipline is measurable: of 830 scored temporal pairs, roughly 719 were suppressed by quality gates and 111 promoted.

10. Sentinel-1 SAR Evidence Layer

The SAR branch is intentionally **independent of RemoteCLIP**. Scope is Sentinel-1 VV/VH **backscatter delta** evidence for GRD / monthly-mosaic products — it is *not* interferometric coherence or InSAR.

Verified real-data validation (see `FINAL_VALIDATION_REPORT.md`):

- Real Planetary Computer `sentinel-1-rtc` GRD products ingested through `onboard_aoi_sar`; linear-to-dB conversion performed exactly once ($10 \cdot \log_{10}$), EPSG:32642 at 10 m.
- Temporal matching by physical tile footprint; SAR observations registered (390 rows) and fused into candidate scoring with quality and temporal-match guards.
- Where SAR is unavailable for a tile, the UI states that SAR evidence is unavailable — no values are fabricated.

11. Temporal Intelligence: Velocity, Acceleration, Storyline

`change/temporal_signature.py` turns sequences of pairwise changes into rates:

- **Velocity** — score per day computed across actual acquisition intervals (`/changes/velocity`);
- **Acceleration** — second-order trend;
- **Signature classification** — each tile is classified as *accelerating*, *steady change*, or *stable/decelerating*;
- **Storyline** (`change/storyline.py`) — classified short-term/seasonal storyline stages per tile;
- **Temporal evolution & series** endpoints expose full per-tile histories for the Analysis Studio.

This extends detection from "what changed" to "how fast", separating sudden events from gradual structural growth.

12. Discovery Clustering

`discovery/clustering.py` groups tiles with cohesive embedding signatures (HDBSCAN-style density grouping over the indexed RemoteCLIP vectors, persisted per tile). Analysts who identify one location of interest can sweep every location with comparable visual/semantic characteristics via `/clusters/{id}/members` galleries, without a new query.

13. AOI Explorer — India-Level Geospatial Map

`frontend/src/components/aoi-explorer/AOIExplorer.tsx` (MapLibre GL):

- A **local India satellite basemap** (bundled raster asset, no tile server, no keys) renders the national view fully offline; a dark fallback style is used if the asset is unavailable.

- **AOI markers** are placed from the catalog's real data (`/aois` returns each AOI's bounding box, tile/scene counts and mosaic thumbnail URL).
- **Hover** shows a preview card (tiles, scenes, observation history span, sensor); **click** flies the map into the AOI's *actual observed footprint* derived from real tile footprints, overlaying the latest mosaic.
- Markers navigate directly into the AOI's detail pages — analysis begins from geography, not from a form. The basemap is a non-authoritative visual backdrop; authoritative geometry always comes from the backend catalog.

14. Qwen Analyst Assistant (On-Device, Grounded)

The assistant (`change/llm_brief.py`, `change/analyst_chat.py`, endpoints `POST /tiles/{id}/analysis/brief` and `/analysis/chat`) runs a local Qwen 2.5 0.5B model through Ollama on CPU:

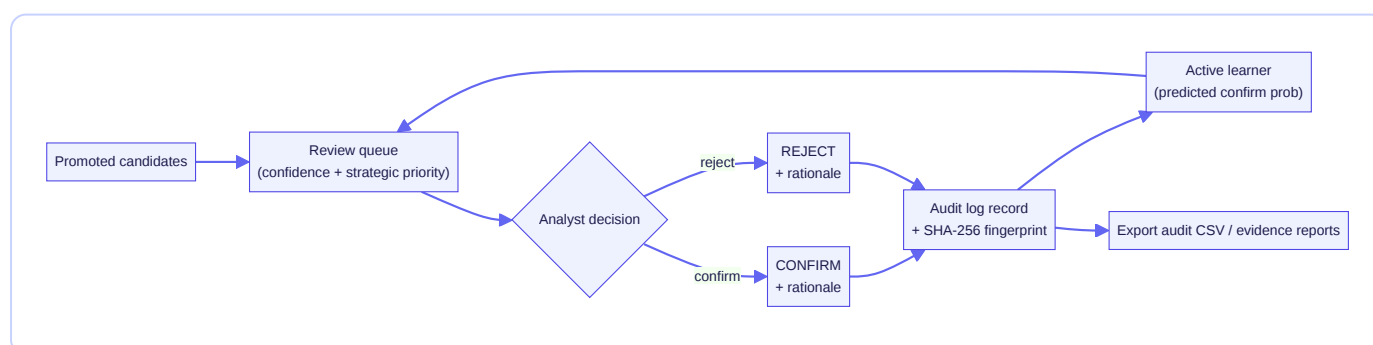
- **Grounding:** the model receives only structured facts already computed by the deterministic system (scores, NDVI/NDWI deltas, SAR status, dates).
- **Guards:** prompts prohibit invented numbers, dates, coordinates, confidence values and operational recommendations; a numeric guard rejects outputs containing unsupported numbers; catalog access is read-only.
- **Honesty by design:** asked whether a change is significant, the assistant answers that the evidence alone cannot establish significance, rather than inventing certainty.
- **Deterministic fallback:** if Ollama or the model is unavailable, the system degrades to a deterministic, rule-based summary — it never guesses.

15. Analyst Evidence Report

`change/evidence_report.py` + `POST /tiles/{id}/analysis/report` generate a one-page dossier locally (PDF via Pillow, or JSON):

- before/after imagery figures, spectral evidence tables (index, before, after, delta), temporal evidence (observation count, interval, velocity, acceleration, trend), SAR status, limitations, and the analyst summary;
- generated on the fly from the local catalog with full source-scene and processing provenance; nothing is uploaded.

16. Analyst Workflow, Active Learning and Audit

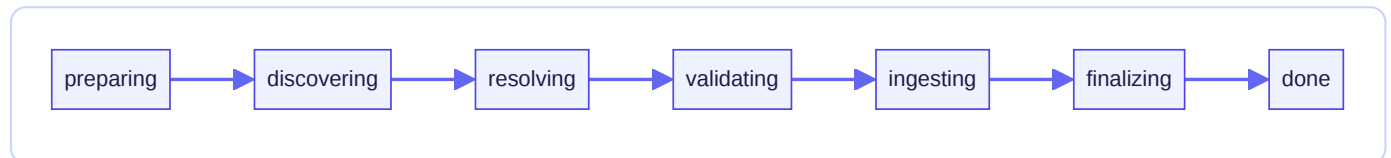


- **Review queue** (`review/queue.py`, `change/priority.py`): triage by confidence and strategic priority (analyst-configured AOI tiers, optional GeoJSON priority zones, proximity to confirmed hotspots). Priority reorders candidates; it never silently drops them.
- **Decisions:** `POST /review/{candidate_id}/decision` with optional analyst rationale; `record_decision` in `catalog_db.py` persists the decision and computes a **SHA-256 record fingerprint** over the decision record.

- **Audit trail:** `GET /review/audit-log` returns every decision with timestamp, rationale and fingerprint; exportable as CSV. Fingerprint verification is read-time and recomputed — verifiable, exportable provenance.
- **Feedback-informed reranking:** analyst decisions feed a lightweight active learner whose predicted-confirm probability reorders the queue as the analyst works.

17. Incremental Onboarding Pipeline

`POST /aois/onboard` starts an **asynchronous job** (in-process job registry with status polling at `GET /aois/onboard/{job_id}`):



- Each stage and each scene emit progress callbacks (`on_stage` , `on_scene_done`); the UI shows a live scene table (accession, date, status) with per-scene outcomes including `skipped_duplicate` .
- **Deduplication:** scenes already in the catalog are recognized and skipped — re-ingestion never double-counts (verified live: a demo folder of three scenes onboards end-to-end in seconds with all three skipped as duplicates).
- **No rebuild:** tiles are embedded and appended to the FAISS index and catalog without reprocessing the archive.
- A **live onboarding demo** panel drives the same real pipeline with a demo folder, proving (not claiming) incremental ingestion.

18. API Surface (verified endpoints)

Group	Endpoints
System	GET /health, GET /stats, GET /system/llm-status
Catalog & AOs	GET /aois, GET /aois/{id}/tiles, GET /aois/{id}/timeline, GET /aois/{id}/mosaic, GET /aois/{id}/narrative
Tiles & analysis	GET /tiles/{id}, GET /tiles/{id}/observations, GET /tiles/{id}/analysis, GET /tiles/{id}/analysis/series, GET /tiles/{id}/similar, GET /vectors/{id}/similar, GET /tiles/compare
Temporal intelligence	GET /changes/velocity, GET /tiles/{id}/temporal-signature, GET /tiles/{id}/temporal-evolution, GET /tiles/{id}/storyline
Search	POST /search/text, POST /search/image, POST /preview/image
Change analysis	GET /changes/candidates, GET /changes/candidates/data-quality, GET /changes/priority, POST /changes/recompute-priority, GET /changes/{id}, GET /changes/{id}/explanations, POST /changes/{id}/brief
Calibration	POST /calibration/run, GET /calibration/results
Review & audit	POST /review/{id}/decision, GET /review/audit-log, GET /export
AI assistant & reports	POST /tiles/{id}/analysis/brief, POST /tiles/{id}/analysis/chat, POST /tiles/{id}/analysis/report
Clustering	GET /clusters, GET /clusters/{id}/members
Onboarding	POST /aois/onboard, GET /aois/onboard/{job_id}
SAR	POST /sar/observations, GET /sar/observations

19. Data Model (logical entities)

Entity	Key information
aois	name/slug, bounding box (derived from real tile footprints), tile/scene counts, priority tier
scenes	source path, sensor, acquisition date, Copernicus metadata, chips extracted
tiles	MGRS-consistent spatial identity, source path, dates, spectral features, land-cover context, temporal summary fields
vector index entries	512-d RemoteCLIP embeddings, ID-mapped to tiles
change_candidates	evidence scores, modality, SAR/fusion values, priority, classification, heatmap/mask artifacts, storyline
decisions / audit log	candidate, decision, analyst rationale, timestamp, SHA-256 record fingerprint
sar_tiles / SAR observations	VV/VH dB features, sensor/product provenance, temporal match info
onboarding jobs	job id, AOI, stage, per-scene outcomes (ingested / skipped_duplicate / failed)

Migrations are additive; new fields are optional so older frontend or database paths do not break.

20. Frontend Architecture

React 18 + Vite + TypeScript SPA (Tailwind CSS, Radix UI primitives, TanStack Query for data fetching with polling, Recharts for signal charts, MapLibre GL for the AOI Explorer). Verified pages:

Page	Purpose
SignInPage	mission-console entry
DashboardPage	live catalog & surveillance stats (all values from /stats , never hardcoded)
A0IPage + A0IExplorer	India-level satellite map, AOI markers, fly-to real footprints
A0IDetailPage	scene ingestion & provenance timeline, mosaic, per-AOI tiles
SearchPage	text & image-chip semantic search with filters and result comparison
ChangesPage , ChangeDetailPage	candidate surveillance grid & full evidence chain (heatmap → mask → SAR → velocity)
VelocityPage	change velocity monitor, temporal activity matrix, tile history, compare-any-two-dates
AnalysisStudioPage	any-two-date analysis, signal charts, analyst remark, Qwen assistant, evidence report export
TileDetailPage	tile spectral signature, nearest visual neighbours, provenance
ReviewQueuePage	human-in-the-loop triage with confirm/reject + rationale
ClustersPage , ClusterDetailPage	discovery clustering galleries
OnboardPage	incremental onboarding form + live demo panel with staged progress
AuditLogPage	decision audit trail with SHA-256 fingerprints, CSV export
SettingsPage	backend/model inspection; offline execution mode: enforced (no external LLM / cloud calls)

21. Deployment and Sovereignty

- **Packaging:** single Docker container (backend + built frontend served by FastAPI); `docker-compose.yml` mounts `data/` and `models/` externally.
- **Offline enforcement:** the settings page surfaces the runtime contract — API base local, offline execution enforced, Ollama reachable locally, no external LLM/cloud calls.
- **Hardware:** CPU-only; no GPU required. Measured latencies and hardware are recorded in `docs/EVALUATION.md` and `PERFORMANCE_REPORT.md`.
- **Scale path:** SQLite → PostgreSQL + PostGIS; FAISS flat → IVF/HNSW for approximate search at national scale; ingestion scales through the same incremental onboarding pipeline.

22. Validation State

Evidence in `FINAL_VALIDATION_REPORT.md` and `PERFORMANCE_REPORT.md`:

- Python regression suite passing (pytest, including focused velocity, LLM, discovery and SAR tests); production frontend build (Vite/TypeScript) succeeds.
- Playwright browser suite covers all declared routes, reload and back/forward navigation, rendered content, console/page errors, HTTP 5xx, the SAR-backed candidate workflow, and the dashboard → AOI → search → tile → velocity → change → discovery flow.

- Real Sentinel-1 GRD ingestion and optical/SAR fusion verified on real products, including the SAR rescue-guard behaviour (low-scoring SAR does not bypass optical quality suppression).
- On-device Qwen verified end-to-end with numeric and domain-grounding guards active; deterministic fallback tested.
- Dashboard values are rendered from `/stats` — nothing is hardcoded.

23. Known Limitations and Scope Honesty

- Zero-shot change-type classification is a candidate interpretation, not ground truth.
- Land-cover classification is heuristic (spectral context), not a supervised land-cover product.
- SAR backscatter delta is not interferometric coherence and is not described as such; fuller SAR fusion awaits value-range and co-registration validation.
- MGRS tiling provides spatial grouping but does not guarantee pixel-level cross-sensor co-registration; SAR matching is by physical tile footprint and temporal compatibility.
- Semantic search uses an RGB-like representation; the multispectral cube is reserved for physical evidence and quality analysis.
- The assistant reports measured facts; it explicitly declines to establish significance or causation the evidence cannot support.

24. Demonstration Flow (as recorded)

Dashboard → AOI Explorer (India map → fly-to real footprint) → tile spectral signature & provenance → semantic text search ("water body expansion") → image-chip search → change candidate evidence chain (heatmap → mask → SAR → velocity) → human review with reject + rationale → velocity monitor & tile history → Analysis Studio (any two dates, signals) → analyst remark → Qwen assistant (summarize; "is it significant?" → honest refusal) → evidence report export → clusters → live incremental onboarding (stages, duplicates skipped) → audit log with SHA-256 fingerprints → settings (offline enforced).

GeoSpectra — semantic retrieval, multi-temporal change analysis, human-verified evidence: completely offline, on commodity hardware.